

G-CSF (Granulocyte-colony stimulating factor)

G-CSF is a type of protein called a growth factor. It is used to increase the number of white blood cells in the blood.

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What is G-CSF?

G-CSF (granulocyte-colony stimulating factor) is a type of protein called a growth factor.

G-CSF stimulates the bone marrow to make more blood cells, and increases the number of some types of [white blood cells](#) in the blood. It can be used to increase white blood cells during [chemotherapy](#) treatment cycles. It can also be used before and after a [stem cell transplant](#).

G-CSF is made naturally in the body, but it can also be made as a drug. Treatments include:

- lenograstim (Granocyte[®])
- filgrastim (Neupogen[®], Nivestim[®], and Zarzio[®])
- lipegfilgrastim (Lonquex[®])
- pegfilgrastim (Neulasta[®] and Pelmeg[®]).

It is best to read this information with our general information about the [type of cancer](#) you have and the [treatment](#) you are having.

Your cancer team will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

When G-CSF is used

Chemotherapy

G-CSF is often given during cycles of [chemotherapy](#). treatment. Chemotherapy can reduce the number of white blood cells in your blood. These cells fight infection. If the number of white blood cells is low, you are more likely to get an [infection](#). A low white blood cell count is sometimes called neutropenia.

The number of white blood cells usually goes back up naturally between chemotherapy cycles. Some people might be given G-CSF to help increase their level of white blood cells between treatments.

Other people may be given G-CSF to reduce the risk of infection when their white blood cells are low

Collecting stem cells for treatment

Stem cells are blood cells at the earliest stage of their development. A [stem cell transplant](#) is sometimes used to treat [blood cancers](#).

You have a course of G-CSF injections for a few days. This stimulates your bone marrow to make more stem cells. The stem cells can then be collected, stored and given when the transplant is needed.

How G-CSF is given

You usually have G-CSF as an injection under the skin (subcutaneously). It is usually given into your thigh or tummy area. You, or someone who looks after you, may be shown how to give the injections. Or they can be given by a nurse at home or at the GP surgery.

If you miss a dose, contact the hospital on the 24-hour number for advice.

Some people have G-CSF as a drip into a vein, which takes about 30 minutes. You have this at the hospital or clinic.

During your treatment, you will meet someone from your cancer team, such as a:

- cancer doctor
- specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

You will have regular blood tests to check the number of white blood cells in your body. Your doctor, nurse or pharmacist will explain how often you need the injections.

Injection site

The skin in the area that was injected may become:

- painful
- swollen
- red – if you have black or brown skin, this can be harder to notice, but the skin might become darker

If this happens, let your doctor know. Painkillers may help.

Storing G-CSF

Follow the storage instructions your pharmacist gives you.

Some G-CSF injections should be stored in the fridge. Take the injection out of the fridge 30 minutes before you use it, so it can come up to room temperature. Your pharmacist will tell you how to dispose of your used needles.

Keep your injections out of sight and reach of children. You should return any unused injections to the hospital.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Side effects while treatment is being given

Some people may have side effects while they are being given this treatment or shortly after they have it.

Allergic reaction

Some people have an allergic reaction while having this treatment. Signs of a reaction can include:

- feeling hot or flushed
- shivering
- itching
- a skin rash
- feeling dizzy or sick
- a headache
- feeling breathless or wheezy

- swelling of your face or mouth
- pain in your back, tummy or chest.

If you are having your treatment in hospital, your nurse will check you for signs of a reaction during your treatment. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

If you are having your treatment at home and develop any of these signs or feel unwell, contact the hospital straight away on the 24-hour number.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

Bone pain

This is caused by your bone marrow making blood cells. You might feel a dull ache in your pelvis (between your hips), your back, arms or legs. Tell your doctor or nurse if you have this. They may suggest medicines to help.

Headaches

This treatment may cause headaches. If you have headaches, tell your doctor, nurse or pharmacist. They can give you advice about painkillers that may help. Tell them if the headache does not get better, or gets worse.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- blood when you cough
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Risk of infection

GCSF is often given with chemotherapy. Chemotherapy can reduce the number of white blood cells in your blood.

An infection can be very serious when the number of white blood cells is low. It is important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

It is important to follow any specific advice your cancer treatment team gives you.

Feeling sick

Your doctor, nurse or pharmacist may prescribe anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to, even if you do not feel sick. It is easier to prevent sickness than to treat it after it has started.

If you feel sick take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids. If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice. They may change your anti-sickness treatment. Let them know if you still feel sick.

Diarrhoea

This treatment may cause [diarrhoea](#). Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

- taking anti-diarrhoea medicines
- drinking enough fluids to keep you hydrated and to replace lost salts and minerals
- any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

Effects on the liver

This treatment can affect how the liver works. You will have regular blood tests to check this. Sometimes liver changes can be serious. Contact the hospital straight away on the 24-hour number if you have any of these symptoms:

- yellow skin or eyes
- feeling very sleepy
- dark urine (pee)
- unexplained bleeding or bruising
- pain in the right side of your tummy (abdomen).

If you have black or brown skin, changes in your skin colour may be less noticeable. You might first notice the whites of your eyes look yellow, or that the colour of your urine has changed.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Constipation

This treatment can cause [constipation](#). Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- [Eat high-fibre foods](#), such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

Loss of appetite

This treatment can affect your appetite. Don't worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

Effects on the kidneys

This treatment can affect how the kidneys work. This is usually mild and goes back to normal after treatment ends. You will have blood tests to check how well your kidneys are working. Contact the hospital on the 24-hour number if you:

- have blood in your urine (pee)
- are passing less urine or peeing less often than usual.

Drinking fluids helps protect your kidneys. The advice is usually to try to drink at least 2 litres (3½ pints) of fluid each day. But follow any advice from your doctor, nurse or pharmacist about how much is right for you.

Effects on the spleen

The spleen is on the left side of the tummy (abdomen), just under the ribs. The spleen helps fight infection and also stores and breaks down blood cells.

G-CSF can cause your spleen to grow bigger. Symptoms of an enlarged spleen include discomfort on the left side of the tummy or shoulder, and feeling full quickly when eating. This is usually mild and goes away after you stop treatment.

If you notice this, contact the hospital straight away on the 24-hour number.

Leaking from tiny blood vessels (capillary leak syndrome)

Rarely, this treatment can cause fluid to leak from tiny blood vessels called capillaries. This can cause low blood pressure and make you very unwell.

The first symptoms may be similar to a head cold with a runny nose. Contact your doctor straight away if you:

- feel faint or dizzy
- feel sick and have diarrhoea
- feel breathless
- have aching or swollen legs.

Effects on the lungs

This treatment can cause changes to the lungs. Tell your doctor, nurse or pharmacist if you develop the following symptoms:

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- a cough that does not go away
- coughing up blood
- wheezing
- breathlessness.

You should also tell them if any existing breathing problems get worse. You may have tests to check your lungs.

Skin changes

This treatment can affect your skin. It might feel dry. You may develop a rash, which may be itchy.

Always tell your doctor, nurse or pharmacist about any [skin changes](#). They can give you advice or prescribe creams or medicines to help.

If your skin feels dry, try using soap-free cleansers and unperfumed moisturising cream every day.

Serious skin changes

Rarely, this treatment can cause a serious skin reaction that needs to be treated immediately in hospital. Contact the hospital straight away on the 24-hour number if you have any of these symptoms:

- a skin rash that is spreading
- blistering or peeling skin
- flu-like symptoms, such as a high temperature and joint pain
- sores on your lips or in your mouth.

Dizziness

This treatment may cause you to feel dizzy (light-headed). If you notice this, do not drive or operate any machinery

Difficulty sleeping (insomnia)

Sometimes this treatment can affect your sleep. Tell your doctor, nurse or pharmacist if cancer treatment makes it [difficult to sleep](#).

Blood pressure changes

This treatment can affect your blood pressure. Tell your cancer team if you have had any problems with your blood pressure. They can monitor this if needed.

Effects on the blood vessels

Very rarely, G-CSF can cause inflammation in a large blood vessel called the aorta. The aorta goes from your heart down through your tummy area (abdomen). Contact the hospital straight away on the 24-hour number if you have any of these symptoms:

- fever
-

pain in the tummy area and back.

Other information

G-CSF is a growth factor. This means that in some people, it may cause cancer to progress or grow. Your cancer team will consider the benefits and risks of this treatment before deciding whether it is suitable for you. They will talk to you about this and answer any questions you have.

Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Sorbitol

Some G-CSF injections contain a sugar called sorbitol. If you have an intolerance to some sugars, talk to your doctor, nurse or pharmacist before taking G-CSF.

Contraception

Your doctor, nurse or pharmacist will advise you not to [get pregnant](#) or [make someone pregnant](#) while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

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Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



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